Experiment 20: Information Retrieval System using TF-IDF in Python

1. Aim

To write a Python program to implement a basic information retrieval system using TF-IDF for document ranking.

2. Procedure

1. Import `TfidfVectorizer` and `cosine_similarity` from `sklearn`.
2. Define a corpus of documents and a search query string.
3. Transform the documents and query into TF-IDF vectors using `TfidfVectorizer`.
4. Compute cosine similarity scores between the query and all documents.
5. Display the similarity scores and the highest-ranked document.

3. Program

```
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity

docs = ["Python is used for NLP.", "NLP processes human language.", "Java is a language."]
vec = TfidfVectorizer()

tfidf = vec.fit_transform(docs)
scores = cosine_similarity(vec.transform(["Python NLP"]), tfidf)[0]
print("Document Scores:", scores.round(2))
print("Top Ranked Document:", docs[scores.argmax()])
```

4. Output

```
Document Scores: [0.68 0.25 0.  ]
Top Ranked Document: Python is used for NLP.
```

5. Result

Thus, the Python program was successfully executed to implement a basic information retrieval system using TF-IDF for document ranking.